

Solutions of the Exercise Problems of Chapter 2

PHY209-Fall2020

Problem 2.14

Show that linear momentum operator $\hat{p}_x$ and the Hamiltonian operator $\hat{H}$ commute each other, when the potential energy V is a constant.

Solution

$$\begin{aligned}
 [\hat{p}_x, \hat{H}] &= [\hat{p}_x, \hat{T} + \hat{V}] \\
 &= \left[\hat{p}_x, \frac{\hat{p}^2}{2m} + \hat{V} \right] \\
 &= \left[\hat{p}_x, \frac{\hat{p}^2}{2m} \right] + [\hat{p}_x, \hat{V}] \\
 &= \frac{1}{2m} [\hat{p}_x, \hat{p}_x^2] + \frac{1}{2m} [\hat{p}_x, \hat{p}_y^2] + \frac{1}{2m} [\hat{p}_x, \hat{p}_z^2] + [\hat{p}_x, \hat{V}] \\
 &= 0 + 0 + 0 + (\hat{p}_x \hat{V} - \hat{V} \hat{p}_x) \\
 &= \hat{V} \hat{p}_x - \hat{V} \hat{p}_x \quad [\text{since } \hat{V} \text{ is constant}] \\
 &= 0.
 \end{aligned}$$

Therefore, the linear momentum operator $\hat{p}_x$ and the Hamiltonian operator $\hat{H}$ commute each other.

Problem 2.15

What is the meaning of the phenomenon when two operators $\hat{A}$ and $\hat{B}$ commute each other.

Solution

When two operators $\hat{A}$ and $\hat{B}$ commute each other, it means that the physical observables A and B corresponding to these operators can be measured accurately simultaneously.

Problem 2.16

Show that

$$\left[\hat{x}, \frac{\partial}{\partial x} \right] = -1.$$

Solution

We consider an eigenfunction ψ . Thus,

$$\begin{aligned}
 \left[\hat{x}, \frac{\partial}{\partial x} \right] \psi &= \left(\hat{x} \frac{\partial}{\partial x} - \frac{\partial}{\partial x} \hat{x} \right) \psi \\
 &= x \frac{\partial \psi}{\partial x} - \frac{\partial}{\partial x} (x\psi) \\
 &= x \frac{\partial \psi}{\partial x} - x \frac{\partial \psi}{\partial x} - \psi \\
 &= -\psi.
 \end{aligned}$$

Therefore,

$$\left[\hat{x}, \frac{\partial}{\partial x} \right] = -1.$$

Problem 2.17

Show that

$$\hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} = \hat{x} \frac{d}{dx} - 1.$$

Solution

We consider an eigenfunction ψ , then

$$\left(\hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} \right) \psi = x^2 \frac{d}{dx} \left(\frac{\psi}{x} \right) = x^2 \left(\frac{1}{x} \frac{d\psi}{dx} - \frac{1}{x^2} \psi \right) = \left(x \frac{d\psi}{dx} - \psi \right).$$

Therefore, from the above equation for any ψ , we can write

$$\hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} = \hat{x} \frac{d}{dx} - 1.$$

Problem 2.18

Show that

$$[\hat{p}_x, \hat{V}(x)] = -i\hbar \frac{dV(x)}{dx}.$$

Solution

Let us consider an eigenfunction ψ . WE have

$$\begin{aligned} [\hat{p}_x, \hat{V}(x)]\psi &= (\hat{p}_x \hat{V}(x) - \hat{V}(x) \hat{p}_x)\psi \\ &= \left(-i\hbar \frac{d}{dx} V - V \left(-i\hbar \frac{d}{dx} \right) \right) \psi \\ &= -i\hbar \frac{d}{dx} (V\psi) + V(i\hbar) \frac{d\psi}{dx} \\ &= -i\hbar V \frac{d\psi}{dx} - i\hbar \frac{dV}{dx} \psi + i\hbar V \frac{d\psi}{dx} \\ &= -i\hbar \frac{dV}{dx} \psi. \end{aligned}$$

Therefore, from the above equation for arbitrary ψ , we can write

$$[\hat{p}_x, \hat{V}(x)] = -i\hbar \frac{dV(x)}{dx}.$$

Problem 2.19

What is a Hermitian operator? Show that the linear momentum operator is Hermitian.

Solution

For simplicity, we consider one-dimensional system. In one-dimension, the linear momentum operator is

$$\hat{p}_x = -i\hbar \frac{d}{dx}.$$

Let us consider two state functions ψ and ϕ of the linear momentum operator $\hat{p}_x$. We have

$$\begin{aligned} (\hat{p}_x \psi, \phi) &= \int (\hat{p}_x \psi)^* \phi \, dx \\ &= \int \left(-i\hbar \frac{d\psi}{dx} \right)^* \phi \, dx \\ &= i\hbar \int \frac{d\psi^*}{dx} \phi \, dx \\ &= i\hbar \int \phi \frac{d\psi^*}{dx} \, dx \\ &= i\hbar \left(\phi \psi^* \Big|_{-\infty}^{\infty} - \int \frac{d\phi}{dx} \psi^* \, dx \right) \\ &= i\hbar \left(0 - \int \psi^* \frac{d\phi}{dx} \, dx \right) \\ &= \int \psi^* \left(-i\hbar \frac{d\phi}{dx} \right) \, dx \\ &= \int \psi^* \hat{p}_x \phi \, dx \\ &= (\psi, \hat{p}_x \phi). \end{aligned}$$

Thus, we see that the linear momentum operator $\hat{p}_x = -i\hbar d/dx$ is Hermitian.

Problem 2.20

Show that the Hamiltonian operator is Hermitian.

Solution

We know the Hamiltonian operator is

$$\hat{H} = \hat{T} + \hat{V},$$

where $\hat{T}$ is the kinetic energy operator and $\hat{V}$ is the potential energy operator. For simplicity of presentation, we consider a one-dimensional system and we also consider two eigenfunctions ψ and ϕ of the Hamiltonian or total energy operator $\hat{H}$. We know the kinetic energy operator

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

We have

$$\begin{aligned} (\hat{H}\psi, \phi) &= \int (\hat{H}\psi)^* \phi \, dx \\ &= \int \left[\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V \right) \psi \right]^* \phi \, dx \\ &= -\frac{\hbar^2}{2m} \int \frac{d^2\psi^*}{dx^2} \phi \, dx + \int \psi^* V \phi \, dx \\ &= -\frac{\hbar^2}{2m} \int \phi \frac{d^2\psi^*}{dx^2} \, dx + \int \psi^* V \phi \, dx \\ &= -\left(\frac{\hbar^2}{2m} \right) \left(\left[\phi \frac{d\psi^*}{dx} \right]_{-\infty}^{\infty} - \int \frac{d\phi}{dx} \frac{d\psi^*}{dx} \, dx \right) + \int \psi^* V \phi \, dx \\ &= -\left(\frac{\hbar^2}{2m} \right) \left(0 - \frac{d\phi}{dx} \psi^* \Big|_{-\infty}^{\infty} + \int \frac{d^2\phi}{dx^2} \psi^* \, dx \right) + \int \psi^* V \phi \, dx \\ &= \int \psi^* \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \phi \, dx + \int \psi^* V \phi \, dx \\ &= \int \psi^* \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V \right) \phi \, dx \\ &= \int \psi^* \hat{H} \phi \, dx \\ &= (\psi, \hat{H}\phi). \end{aligned}$$

Therefore, the Hamiltonian operator $\hat{H}$ is Hermitian.
